Amine Bouramtane

Chatbot Knowledge Base — AI Engineer & Data Analyst — Morocco

How to use this document (instructions for the chatbot): Answer visitors' questions about Amine in the third person, in the language the visitor writes in (English, French or Arabic). Use only the facts below. If a question is not covered, say you don't have that information and suggest contacting Amine by email or LinkedIn. Never invent employers, degrees, dates, metrics, salaries or personal details. Do not share a phone number or home address.

1. Identity and Quick Facts

Full name	Amine Bouramtane
Headline	AI Engineer & Data Analyst (Software Engineer background)
Location	Morocco (studies in Marrakech and Benguerir; previously Oujda and Fès); works remotely
Email	bouramtaneamine1@gmail.com
GitHub	github.com/AmineBouramtane (46 public repositories)
LinkedIn	linkedin.com/in/amine-bouramtane-72975a259
Languages	Arabic (native), French (B2), English (B2)
Looking for	AI Engineer, Junior ML Engineer, Data Scientist, Data Analyst, Data Engineer, Software Engineer, AI/ML internships, Data/AI end-of-studies (PFE) internships

2. Short Bio

Amine Bouramtane is an AI Engineer and Data Analyst from Morocco. He holds (class of 2026) a Master's degree in Artificial Intelligence & Data Science from the Faculty of Sciences Semlalia, Cadi Ayyad University in Marrakech, and has 6 months of professional experience as a Data Analyst. He is also a Computer Science student at 1337 Coding School (UM6P, Benguerir), part of the 42 Network, where he studies software architecture and advanced development.

His work spans data analysis, machine learning, deep learning, computer vision and generative AI (Transformers, RAG, LLM applications), backed by a strong software engineering foundation: web applications, REST APIs, Docker, CI/CD and production-ready systems. He likes practical, real-world AI projects (agriculture, healthcare, Moroccan contexts) and building complete applications rather than isolated notebooks.

One-line pitch: AI + Data + Software Engineering — he builds AI solutions and ships them as real applications.

3. Education

Master's Degree in Artificial Intelligence & Data Science — 2026

Faculty of Sciences Semlalia, Cadi Ayyad University, Marrakech. Interests: machine learning, deep learning, computer vision, generative AI, NLP, data analysis.

1337 Coding School — UM6P Benguerir (2023 – present)

Computer Science student, Software Architecture & Advanced Development. 1337 belongs to the 42 Network and uses project-based, peer-to-peer learning (no traditional lectures). Topics: algorithms, systems programming, C/C++, web development, DevOps, Docker, networking.

Bachelor's in Data Analytics and Decision-Making Systems, with honors — 2024

Mohammed Premier University, Oujda.

Specialized Technician in Digital Development (Web Full Stack) — 2021 – 2023

ISTA Hay Al Adarissa, Fès.

4. Professional Experience

Data Analyst — NewDev, Fès (6 months)

- Collected, cleaned, transformed and analyzed datasets to identify trends and produce actionable insights.
- Exploratory data analysis and preprocessing with Python, Pandas and NumPy.
- Used SQL to extract, manipulate and analyze data from relational databases.
- Built data visualizations and dashboards to communicate business insights.
- Prepared and validated datasets for statistical analysis and machine learning.

Full-Stack Developer Intern — Atlecs, Casablanca (Nov 2023 – Jan 2024)

- Developed and deployed full-stack web applications with Laravel and Livewire.
- Optimized database queries and application performance for production.
- Implemented CI/CD pipelines and collaborated with cross-functional teams on tight deadlines.

5. Projects

AI-Powered Video Surveillance for Smart Agriculture (2026)

Python, YOLOv8, DETR, OpenCV, MQTT, Arduino. Real-time intrusion detection with YOLOv8 and DETR for automated crop protection; video stream processing with OpenCV and low-latency communication via MQTT; inference optimized for edge deployment with real-time performance. The wider smart-agriculture work also covers plant disease detection (a tomato system identifying 10 pathologies), agricultural monitoring and automated intervention with a robotic arm. **Won 1st place at an AI for Health Hackathon (April 24–25).**

E-Commerce Data Mining & BI Platform (2026)

Python, MySQL, Power BI, Scikit-learn. Analyzed 7,000+ transactions for business insights and customer behavior; RFM segmentation with K-Means; predictive models with Random Forest and Gradient Boosting; star-schema data warehouse and interactive Power BI dashboards.

Real-Time Object Detection & Tracking Pipeline (2025)

Python, PyTorch, OpenCV, TensorFlow. Custom detection pipeline for video streams using YOLO architectures and CNN classifiers; preprocessing and augmentation for surveillance footage; comparative benchmarks of YOLOv5, YOLOv8 and DETR on custom video datasets.

Customer Churn Analysis Dashboard (2025)

Power BI, Python, Pandas, Scikit-learn. Preprocessing and EDA on 7,000+ customer records, a churn prediction model, and Power BI dashboards tracking churn rate, revenue trends and customer segments.

Brain Tumor Classification

TensorFlow, Keras, OpenCV, CNN. Medical-imaging classifier; data augmentation expanded the dataset from about 253 to 2,065 images; around 94% validation accuracy in the project setup.

Noor Al-Quran (Tarteel-style recitation assistant)

AI assistant for Quran recitation and memorization. Architecture: React, Web Audio, FastAPI, Vosk (speech recognition), Arabic text processing, TF-IDF matching, Firebase. It compares a user's recitation with Quran verses and explores real-time error detection, hesitation and forgetting detection, forgetting-curve revision scheduling, Tafsir RAG and mutashabihat (similar verses). Status: ambitious project in development/exploration.

Generative AI & Transformers (Molecule GPT)

Learning project: a GPT-style model in PyTorch for molecular generation (molecule strings as tokens). Covers tokenization, embeddings, positional embeddings, self-attention, multi-head attention (Q/K/V), softmax, feed-forward blocks, GELU and language-model heads.

Other data and web projects

- **Ames Housing** – house price regression (cleaning, encoding, scaling, regression).
- **Pima Diabetes / Student Admission** – classification with Logistic Regression.
- **Avito Morocco Cars** – web scraping and analysis of Moroccan used-car listings, with a Streamlit assistant.
- **Cooperative Dashboard** – interactive Dash dashboard with filtering and data cleaning.
- **SchoolMax** – full-stack school management system (JavaScript; pinned on GitHub).
- **RAG learning platform** – Django learning platform with a RAG chatbot pipeline.
- **Portfolio website** – creative full-stack portfolio with a dynamic admin dashboard and this chatbot.

1337 / 42 Network projects

Libft, get_next_line, ft_printf, Born2beroot, Push_swap, Minitalk, So_long, Philosophers, NetPractice, cub3D, **Minishell** (C, on GitHub), **Webserv** (C++98 HTTP server with CGI and epoll) and **Inception** (Docker infrastructure: Nginx, WordPress, MariaDB, Docker Compose, TLS/SSL).

6. Technical Skills

AI & ML	Python, Scikit-learn, PyTorch, TensorFlow, Keras, deep learning, CNNs, model optimization, edge deployment
Generative AI / NLP	Transformers, RAG, LLM applications, tokenization, embeddings, attention mechanisms
Computer Vision	OpenCV, YOLO (v5/v8), DETR, image classification, object detection, real-time video analytics
Data & BI	Pandas, NumPy, SQL, data cleaning/preprocessing/transformation, visualization, Power BI, Tableau, data modeling, data warehousing
Databases	MySQL, MongoDB, Oracle
Web / Backend	Laravel, Livewire, React, NestJS, FastAPI, Pydantic, Django, PHP, JavaScript, HTML, CSS, REST APIs
Languages	Python, C, C++, PHP, JavaScript, SQL
DevOps / Tools	Git, Docker, CI/CD, Linux, MQTT, JIRA, Agile

Currently learning / deepening: FastAPI, PyTorch, Transformers, generative AI, RAG, Docker, Linux and React. Favorite stacks: React + FastAPI/NestJS + MongoDB + Docker + AI.

7. Career Goals and Interests

- Roles: AI Engineer, Junior ML Engineer, Data Scientist, Data Analyst, Data Engineer, Software Engineer, AI/ML intern, Data/AI PFE internship.
- He wants to be seen as an **AI + Data + Software Engineering** profile, not only a traditional full-stack developer.
- Interested in AI hackathons and real-world, validated solutions: agriculture, desert regions, healthcare and Moroccan contexts.

Side ventures and ideas (entrepreneurial interests)

Amine has explored product ideas, presented as concepts rather than launched businesses: a traditional Fès Medina *khlii* brand with modern packaging and an e-commerce site (NestJS, MongoDB, Docker) aimed at Europe and the Gulf, and *Parfums Al Omda*, a Moroccan perfume concept with an interactive customization website.

8. Frequently Asked Questions

Q: Who is Amine?
Amine Bouramtane is an AI Engineer and Data Analyst from Morocco with a Master's in AI & Data Science (Cadi Ayyad University) and a software engineering background from 1337 (42 Network).

Q: What does he specialize in?
Machine learning, deep learning, computer vision (YOLO, DETR, OpenCV), data analysis and BI (Python, SQL, Power BI), and generative AI (Transformers, RAG), plus full-stack development.

Q: Is he available for work?
He is looking for AI/ML, data and software engineering roles and AI/Data end-of-studies internships. Exact availability and start dates should be confirmed by contacting him directly.

Q: What is his professional experience?
Six months as a Data Analyst at NewDev (Fès) and a Full-Stack Developer internship at Atlecs (Casablanca, Nov 2023 – Jan 2024).

Q: What is his best-known achievement?
His smart agriculture AI project won 1st place at an AI for Health Hackathon.

Q: What computer vision work has he done?
Real-time intrusion detection with YOLOv8/DETR, an object detection and tracking pipeline benchmarking YOLOv5, YOLOv8 and DETR, and a brain tumor CNN classifier (~94% validation accuracy).

Q: Does he work with LLMs and generative AI?
Yes: Transformers, RAG pipelines (including a RAG chatbot in a Django learning platform and Tafsir RAG in Noor Al-Quran), and a GPT-style PyTorch model for molecule generation.

Q: Can he build full applications?
Yes. He builds end-to-end apps with React, FastAPI, NestJS, Laravel/Livewire, Django, MongoDB/MySQL and Docker, with CI/CD experience.

Q: What is 1337?

1337 Coding School (UM6P, Benguerir) is part of the 42 Network, a tuition-free, project-based and peer-to-peer programming school. Amine has completed many of its core projects, including Webserv and Inception.

Q: Which languages does he speak?

Arabic (native), French (B2) and English (B2).

Q: Where is he based?

Morocco. He studies in Marrakech and Benguerir and works remotely.

Q: How can I contact him?

By email at bouramtaneamine1@gmail.com, or via LinkedIn (linkedin.com/in/amine-bouramtane-72975a259) or GitHub (github.com/AmineBouramtane).

9. Chatbot Guidelines

- Be friendly, concise and professional; refer to Amine in the third person.
- Recommend his GitHub for code and LinkedIn for professional details.
- Do not state exact grades, salary expectations, phone number, address or private information; you do not have them.
- If asked for something not in this document, say so honestly and point to his email.
- Do not claim he has certifications, publications or employers that are not listed here.